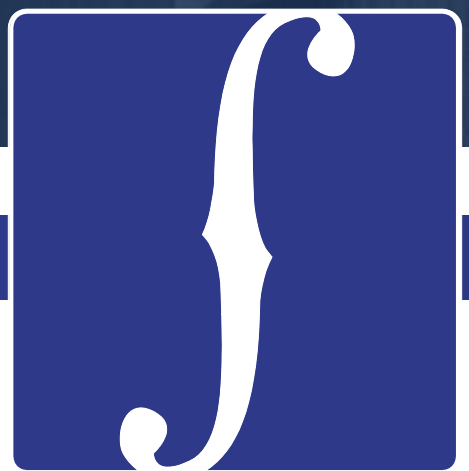




CELLO

Identifying HARPed Loans

January 2016

1

Importance of identifying HARP loans

Loans that have been refinanced through the Home Affordable Refinance Program (HARP) generally prepay slower than other refinance loans that have undergone full underwriting. As a streamline refinance program, HARP generally made only minimal requirements of borrowers, and such borrowers may have trouble qualifying for regular refinances.

2

Existing algorithms for identifying HARPed Loans

1. Prior to August 2013:
 - This is the period prior to Freddie Mac providing loan-level credit information for fully underwritten mortgages
 - MI matches were used to identify potential HARPed loans with OLTV > 80% and loan purpose as Refi-NCO (No Cash-out)
 - Although HARP targets with LTVs > 80%, loans with lower OLTVs are counted as part of HARP because the Agencies essentially suspended their streamline refi programs after the financial crisis and currently rely on the HARP program for all streamline processing
2. After August 2013:
 - New Freddie Mac and Fannie Mae loan-level credit information are used to identify 30-year fixed –rate HARP loans by mapping the credit dataset to the standard database without restrictions on the LTV ratio
 - The credit data set has about a one year lag, which means that recently originated loans cannot be identified as being HARPed or not using the current algorithm

Three Step Mapping

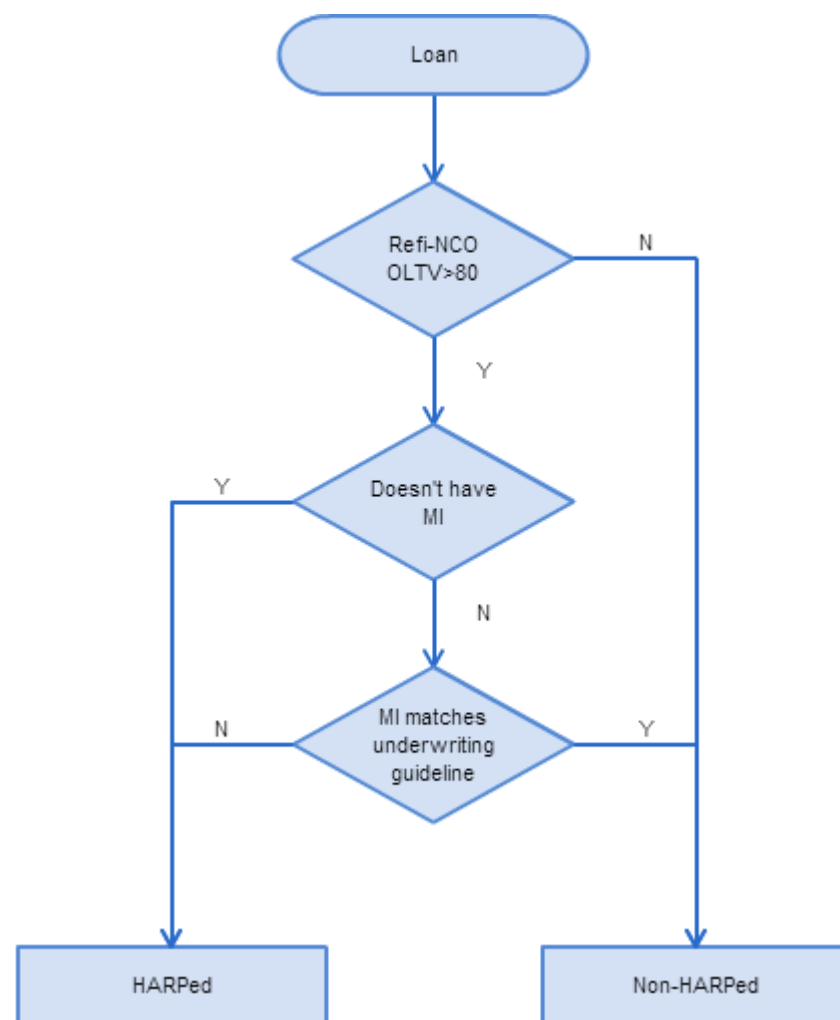
Step 1: The loan should be originated after June 2009 as a refi-NCO loan, with Origination LTV >80%. Otherwise, it's Non-HARPed.

Step 2: If the loan doesn't have MI, we match it as HARPed loan;
If not, go to step 3.

Step 3: For the loans that potentially carry over MI check whether MI matches current underwriting guidelines. If there's no match, then it's a HARPed loan.

Cons

1. Cannot match loans with OLTV equal or lower than 80%.
2. Highly sensitive to the correctness of MI underwriting guidelines.

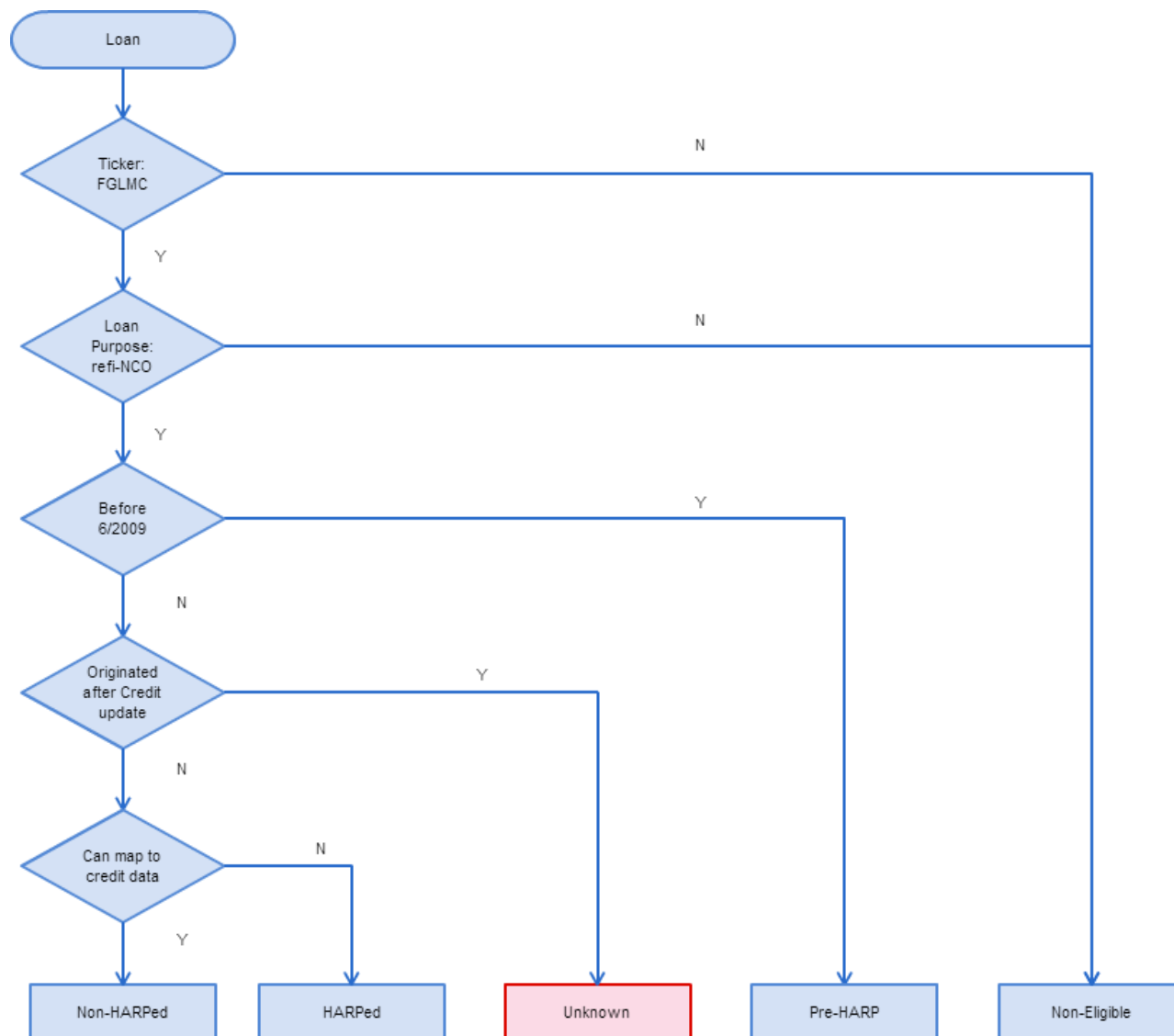


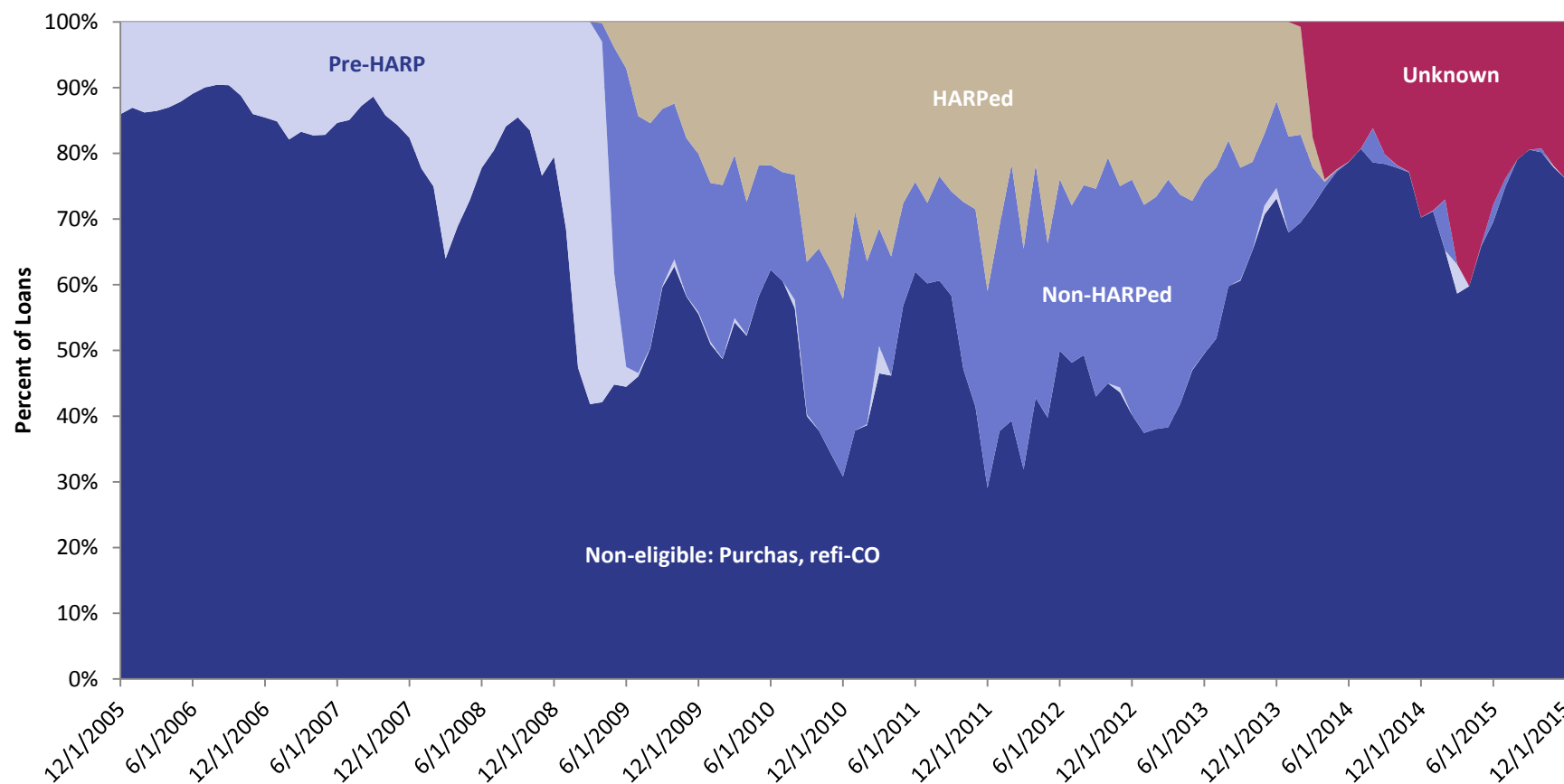
Credit Data Mapping

- No restrictions on OLTV > 80%.
- More accurate as there is no dependency on MI underwriting criteria.

Cons

- The credit data set has about a one year lag, meaning we don't have the most recent origination data. Consequently, we have to mark the HARP status as unknown for recently (within the last year) originated loans.





The blue area can be consider as non-HARPed loans; brown area is HARPed loans percentage.

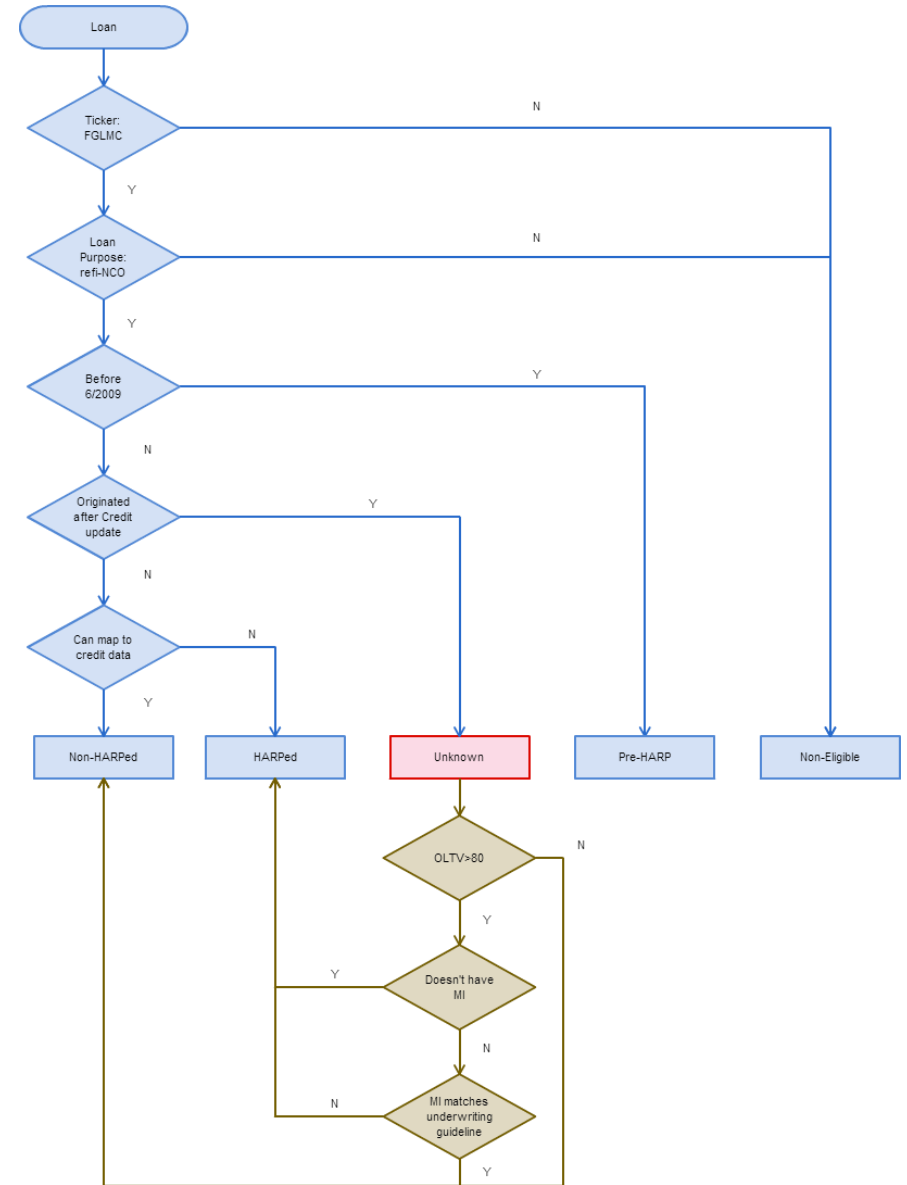
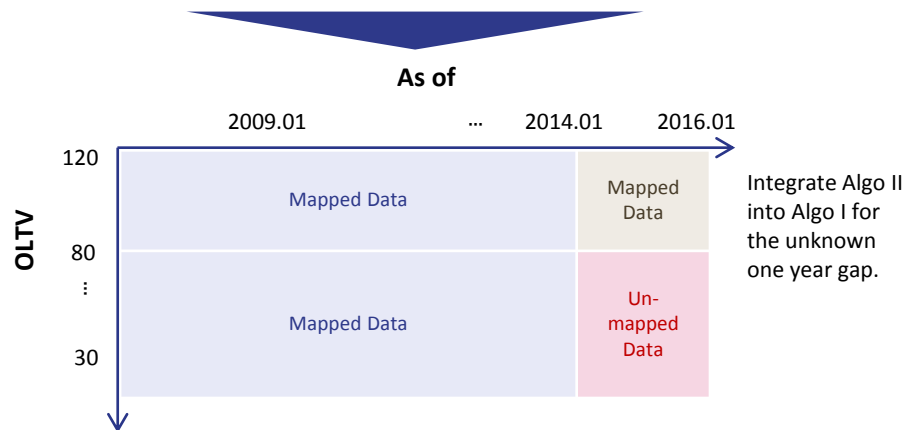
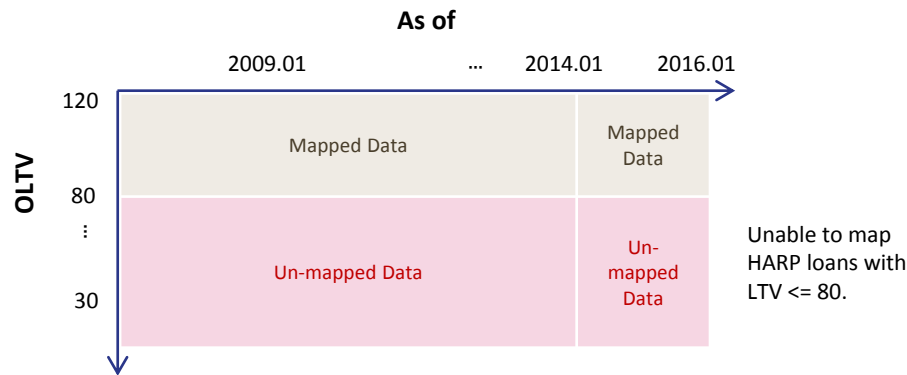
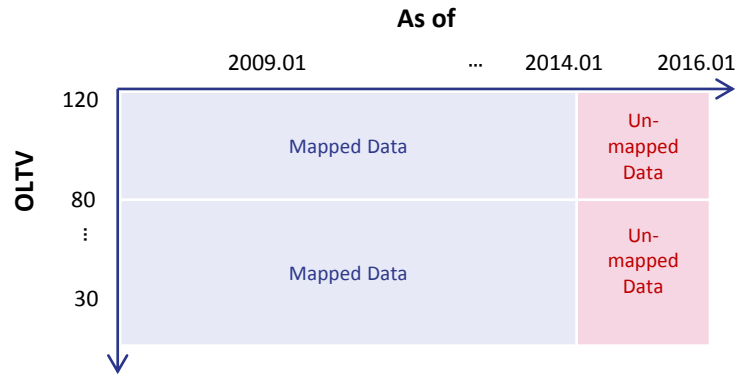
Red area is the loans originated within the most recent year. The credit dataset has a gap on this area.

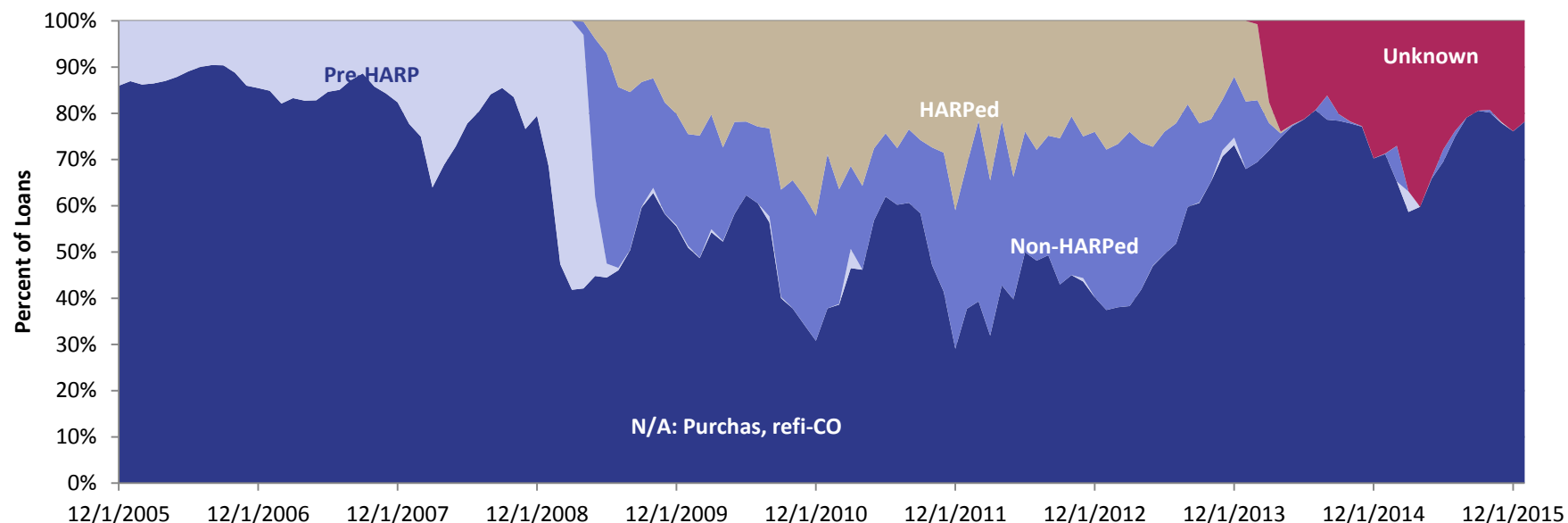
Need to improve the mapping for the red area.

Algorithm I

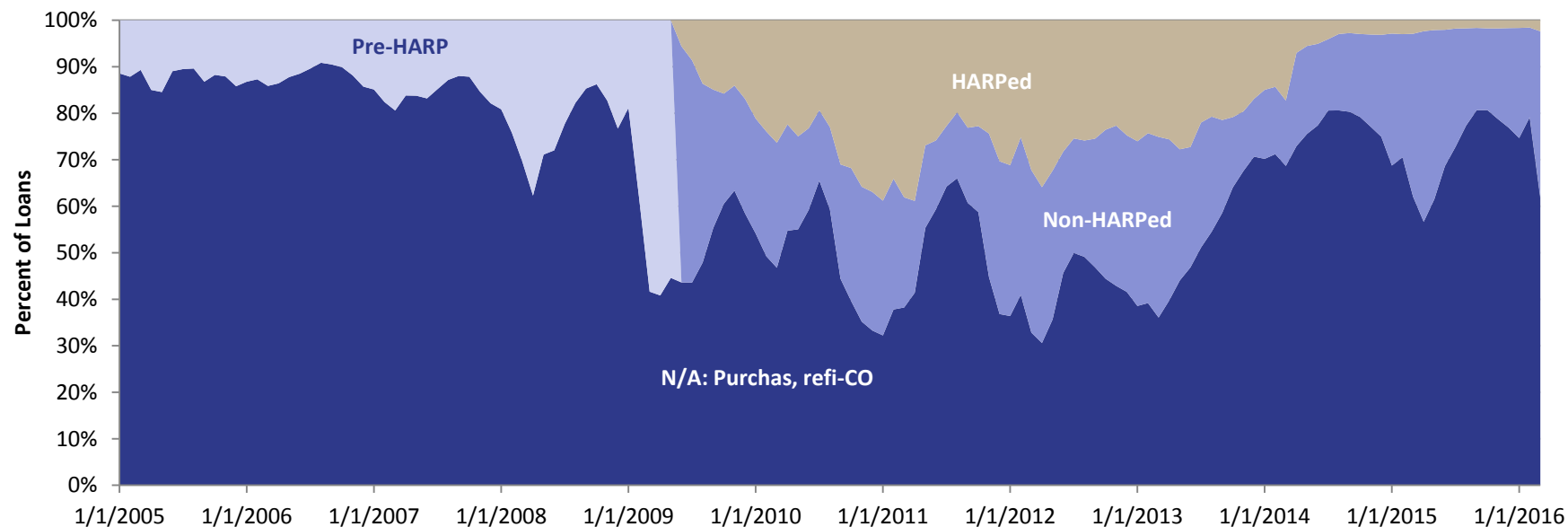
Algorithm II

Enhanced Algo.





Using the Enhanced Algorithm



H1

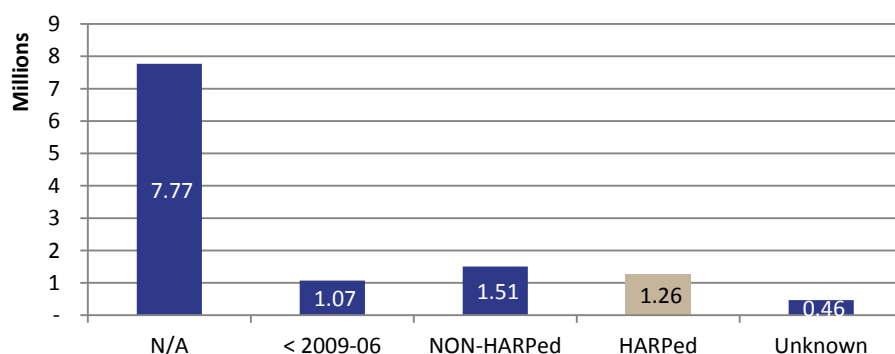
There is a bias between FHLMC/FNMA (FNMA: consider all refis, FHLMC: all refis excluding NCO/CO)

Not true

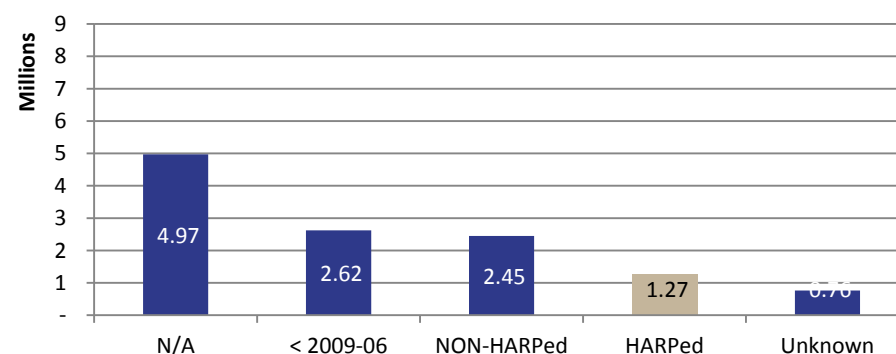
Using FHLMC data, we ran the new algorithm two ways: excluding cash-out refis and considering all refis. The loan count number for HARPed loans were identical under both the old algorithm and new algorithm.

Credit Data Mapping Algo.

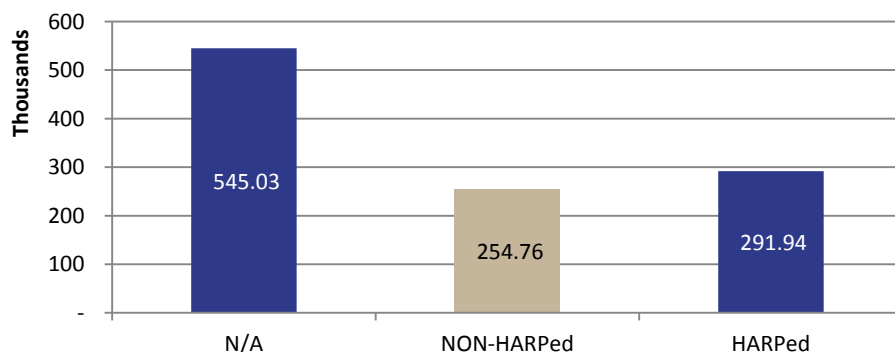
Exclude Cash-out refis (All Periods)



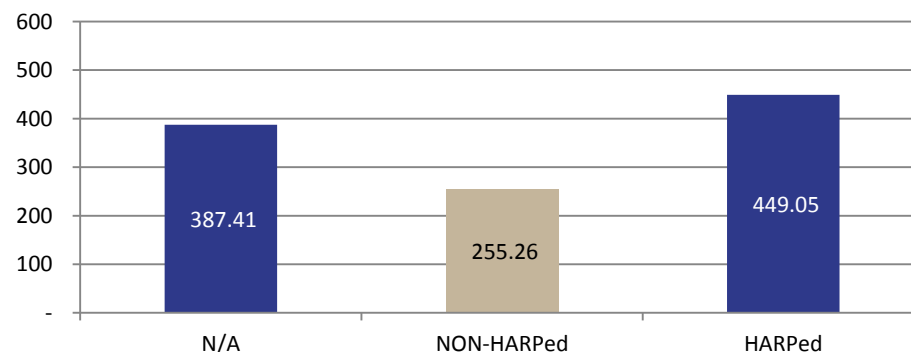
Include Cash-out refis (All Periods)



Exclude Cash-out refis (2013)



Include Cash-out refis (2013)



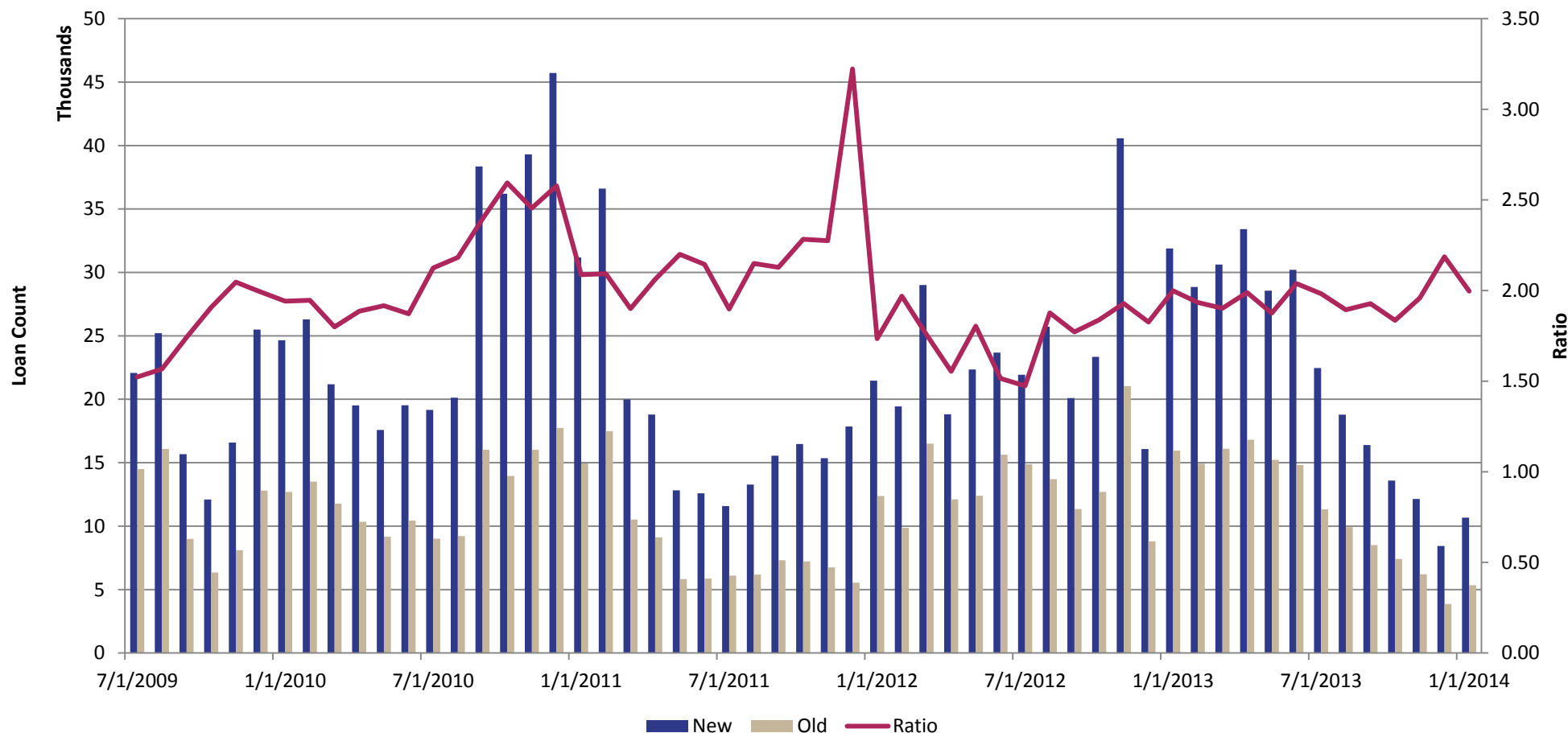
It turns out that all cashout refis are fully documented so the new algorithm provides a non-biased HARPed loan identification process.

H2

There is a bias between the two algorithms

True

By plotting the loan count number for HARPed loans using the two different algorithms, we can see a systematic bias between them.



The **average ratio (New algorithm / Old algorithm)** is 2. So in our enhanced algorithm, we need to adjust the old algorithm by the ratio number for the most recent year period. **Presumably this is because we're not counting loans with LTVs < 80% in the MI algorithm.**